Diffusion Models and Pseudo-Change: A Transfer Learning-Based Change Detection in Remote Sensing Images

Jia-Xin Wang, Teng Li, Si-Bao Chen, *Member, IEEE*, Cheng-Jie Gu, Zhi-Hui You, and Bin Luo, *Senior Member, IEEE*

***Abstract*— Remote sensing (RS) image change detection (CD) has been a research hotspot in recent years, which plays an important role in urban planning and disaster assessment. However, since CD labels are difficult to obtain, how to utilize semantic information in RS images to improve the change prediction performance is a problem worth exploring. To solve this problem, we propose a transfer learning-based CD method that utilizes a diffusion generation model to translate high-level semantic information into low-level change information. First, we propose a pseudo-change image pair generation method that utilizes semantic labels to guide the diffusion model to generate change images. Then, the refined loss (RL) is designed to improve the model's ability to recognize change features based on the difference between pseudo-change image pairs and unlabeled image pairs. Experimental results on WHU-CD, LEVIR-CD, and GoogleGZ-CD datasets show that the proposed method effectively transfers the semantic information into change information and finally improves the model's feature recognition ability for change objects. Compared with recent CD and transfer learning methods, the proposed transfer learning model (TLM) achieves the best performance. The source code is available at https://github.com/VCISwang/STCD.**

***Index Terms*— Change detection (CD), diffusion model, pseudo-label, remote sensing (RS).**

I. Introduction

REMOTE sensing (RS) image processing is an important research component in multimedia processing, and with the development of RS technology, complex feature information is utilized in a variety of methods. For example, building extraction methods [1] detect buildings in the RS images, which plays an important role in urban development planning. Moreover, change detection (CD) techniques [2] based on bi-temporal RS images are applied to capture the change information of the same area at different times. These CD methods can be applied in many fields, such as natural disaster assessment [3] and agricultural land monitoring [4].

Due to the high-resolution and multitemporal characteristics of RS images, traditional CD methods for natural images usually require geometric correction and radiation normalization of the data and then use the thresholding method to obtain rough change prediction. With the advancement of deep neural networks, semantic segmentation and CD methods [5], [6] based on deep learning have achieved great success. The reliability of these deep learning models relies heavily on large-scale data for model training. Nonetheless, the CD task requires extensive resources for the labeling process, as it is difficult to obtain change labels between multitemporal RS images. Therefore, it is worth exploring how to improve the CD performance in the absence of labeled data.

For unlabeled bi-temporal RS images, traditional CD methods use algorithms such as change vector analysis (CVA) [7] and multivariate CD [8] to extract the object information in different time-phase images and then identify the changed areas by comparison. These techniques do not rely on training on labeled data, but the accuracy of prediction results is limited. Recently, researchers have proposed unsupervised CD methods [9] based on deep learning that use deep neural networks to more accurately identify object features. These approaches enable better comparison of target features in images. However, these unsupervised models have problems in effectively discriminating information about changes of interest, such as buildings and roads, due to the lack of labeled data to guide training.

Obtaining single-temporal semantic labels from RS image data is easier than obtaining bi-temporal change labels in a CD task. The advancement of RS technology has provided more and more rich RS image semantic segmentation datasets, such as the WHU building dataset [4], [10], the Inria aerial image labeling dataset [11], and the ISPRS Vaihingen and Postdam datasets [12], to develop methods such as building extraction. In transfer learning-based approaches, Zunair et al. [13] proposed to use single-temporal image semantic tagging to guide model training for CD, combining change prediction with bi-temporal semantic label relationships to

Received 31 July 2024; revised 25 September 2024; accepted 18 October 2024. Date of publication 22 October 2024; date of current version 1 November 2024. This work was supported in part by the Fundamental Research Funds of the Anhui University of Science and Technology (AUST) under Grant 2024JBQN0025, in part by the Natural Science Research Project of Anhui Educational Committee under Grant 2024AH050357, in part by the Scientific Research Foundation for High-level Talents of Anhui University of Science and Technology under Grant YJ20240003, in part by the NSFC Key Project of International (Regional) Cooperation and Exchanges under Grant 61860206004, in part by the NSFC Key Project of Joint Fund for Enterprise Innovation and Development under Grant U20B2068, and in part by the National Natural Science Foundation of China under Grant 61976004. *(Corresponding author: Teng Li.)*

Jia-Xin Wang and Cheng-Jie Gu are with the School of Public Security and Emergency, Anhui University of Science and Technology, Hefei 231131, China (e-mail: jxwang@aust.edu.cn; cjgu@aust.edu.cn).

Teng Li is with the Key Laboratory of Intelligent Computing and Signal Processing, Ministry of Education, Anhui University, Hefei 230093, China (e-mail: tenglwy@gmail.com).

Si-Bao Chen, Zhi-Hui You, and Bin Luo are with Anhui Provincial Key Laboratory of Multimodal Cognitive Computation, MOE Key Laboratory of ICSP, School of Computer Science and Technology, Anhui University, Hefei 230601, China (e-mail: sbchen@ahu.edu.cn; ZhihuiYou@stu.ahu.edu.cn; luobin@ahu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3484526

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

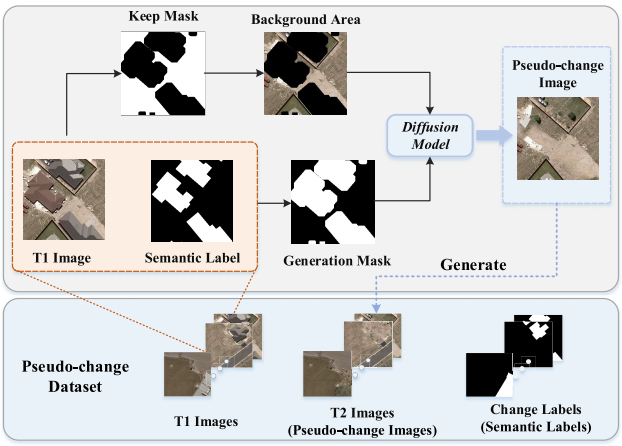


Fig. 1. Generation of the pseudo-change module (GPM). This module utilizes the diffusion model to generate pseudo-change images of specified change regions from RS semantic segmentation images and then uses pseudo-change image pairs to train CD models.

enable the model to better recognize change regions. Recently, Cao and Huang [14] proposed a multistage transfer learning method based on the semantic information from RS images. The method trained a building extraction network using the semantic information from RS and then used the model to segment RS images at different times and compare them to obtain the pseudo-labels. Finally, the method fine-tuned the model to achieve CD. This approach effectively utilizes RS semantic information to obtain a CD model. However, the multitemporal attributes of RS images cause this approach to disregard the connection between building semantic information and changing features.

In this work, we propose a transfer learning CD method based on the denoising diffusion model [15]. The method is inspired by image inpainting algorithms [16], which generate the occluded regions of the original image. Considering the properties of the CD task, we find that it is possible to create pseudo-change bi-temporal images by restoring the building areas in the building extraction datasets to the background. For the proposed generated method, we present some pseudo-change image pairs in Fig. 1. These examples illustrate the success of the denoising diffusion probabilistic model (DDPM) [17], as the pseudo-change pairs are challenging to differentiate from actual bi-temporal change images with real changes. Therefore, in the proposed transfer learning method, we use the diffusion model to generate pseudo-change image pairs to train CD models. The CD network is trained using pseudo-change image pairs to acquire the pretrained model. In addition, since the segmentation dataset and the CD dataset are under different domains, we use the style transfer network [18] to transfer the image domain under the CD task to the pretraining data. After generating pseudo-labels for unlabeled CD datasets using the pretrained model, we propose the refined loss (RL) for real change image pairs and pseudo-change image pairs in the fine-tuning stage to improve the model's ability to detect changed regions.

In summary, the main contributions of this work are as follows.

1) We propose a transfer learning-based approach for building CD, a framework that utilizes only single-temporal phase semantic labels to train the model.
2) The generation of the pseudo-change module (GPM) uses a diffusion model to regenerate the building area, and the pseudo-change image pairs obtained are used for model pretraining.
3) RL is designed based on the difference between the real change area and the pseudo-change areas, which makes the change area in the unlabeled image better discriminated.

The article is organized as follows: Section II describes the related CD methods and compares their advantages and limitations. Section III introduces the proposed transfer learning method. Sections IV and V analyze and discuss the experimental results, as well as explore the algorithm's performance using ablation experiments, to demonstrate the effectiveness of the proposed approach on various datasets. Finally, Section VI summarizes the article.

II. Related Work

In this article, we focus on exploring new approaches using generative models on the task of CD in RS images. Therefore, in this section, we first introduce the research methodology related to CD. Then, for the generative model involved in the approach, the related research is also presented in detail. Finally, we introduce the transfer learning methods that are most relevant to the approach proposed in this article.

A. Change Detection

With the successful application of deep learning to computer vision, there has been a rapid development of object-based CD methods [5], [19], [20] for high-resolution RS images. Daudt et al. [5] proposed a CD method based on the deep learning model that uses a two-stream network to fuse and compare image features from different times. Different fusion methods have explored the diversity of CD networks. Moreover, some CD networks have been proposed based on the UNet [6] structure and have obtained improved results. In the early methods, regarding the data fusion technique, Zhang et al. [21] proposed a multimatrix learning algorithm using multisensor data, which has better results for joint classification. Similar to some semi-supervised methods, Yan et al. [22] designed a Bayesian neural network to calculate uncertainty for high-dynamic-range image processing.

In recent years, researchers have introduced various methods [19], [20] using transform networks to extract multitemporal RS image features for CD with more applications of transform networks [23] in computer vision. To recognize multiscale change targets in CD, Zhang et al. [24] proposed a network structure with multilevel feature fusion. To optimize the utilization of the relationship between local and global features, numerous feature fusion strategies have been proposed by researchers for CD [25], [26], [27].

As deep learning-based supervised CD models often require a significant amount of labeled data to be available, researchers have proposed some semi-supervised and unsupervised CD

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

methods [28], [29], [30]. With reference to the conventional CVA algorithm, Saha et al. [31] employed deep neural networks for extracting semantic information from RS images. The extracted information is then compared to obtain change predictions. Recently, Bandara and Patel [9] proposed an unsupervised method for CD that utilizes a deep-change probability generator and a deep-feature extractor. Besides, Huang et al. [32] proposed a spatiotemporal enhancement and interlevel fusion network (SEIFNet) that improves the feature extraction ability for changing objects. In our experiments, we also include the comparison of these latest CD models.

B. Generative Approaches

In unsupervised methods, it is difficult to obtain reliable predictions due to the lack of supervised information. Applying generative adversarial networks [33], Peng et al. [34] introduced a semi-supervised approach for CD in high-resolution RS images. Their method utilizes a limited number of labeled data and a discriminative network to improve the reliability of pseudo-labels. With the successful application of DDPMs [35] in the field of image generation, more and more research has made progress based on generative models. In the image denoising task using the diffusion model, Yan et al. [36] combined the feature learning ability of the diffusion model to alleviate the ghosting artifacts and other problems of high-dynamic-range images. Based on the structural design of the diffusion model, Nichol and Dhariwal [15] proposed a more efficient model for the image generation method. Some recent studies based on the image generation capability of the diffusion model propose methods for painterly image harmonization [37]. Similarly, when the data is not easy to obtain, Han et al. [38] proposed to utilize the diffusion model to generate a dataset of food images. The method proposed in this article is to generate a CD dataset using semantic labels.

In the proposed transfer learning approach, semantic segmentation datasets are usually under different domains than CD datasets. We apply a style transfer model [18] to align the CD data with the pretrained model for segmentation data. The style transfer approach has also been applied to the design of reconstructive loss [39], which improves the performance of unsupervised CD models.

C. Transfer Learning-Based CD

Because CD in RS images is also a semantic information analysis of multitemporal images, [40] trained the CD model by combining semantic labels while having change labels. The results also show that semantic information can significantly improve CD accuracy. Liu et al. [41], [42] pretrained the model on the RS image segmentation dataset and then extracted features for bi-temporal images, respectively. Finally, change predictions are obtained by calculating the difference between features. Similarly, Saha et al. [43] pretrained the model on the image classification dataset and then grouped the predictions for comparison on multispectral RS data. In a recent study, Cao and Huang [14] proposed a multitask network for simultaneous RS image segmentation and CD and optimized the pseudo-label generation method. This method applies RS semantic information transfer to the CD task.

The proposed method differs from other transfer learning approaches in that we try to use the semantic information of RS images to obtain pseudo-change image pairs for CD. The recent face repainting method [16] can obtain high-quality images by using the diffusion model [15]. In some research on improving CD performance using single-temporal semantic labels, Chen et al. [44] proposed to divide single-temporal images into different blocks for exchange to create change information. This method is similar to the method proposed in this article, but due to the powerful image generation ability of the diffusion model, the proposed pseudo-change information is relatively more realistic and effective. The proposed method in this article, inspired by these methods, uses the denoising diffusion model [17] to generate a part of the background by erasing the semantic objects in RS images and then saving it as a pseudo-change image. For image segmentation and CD tasks, the proposed method is improved based on the multitask network proposed by [14] and then combined with pseudo-change images for training. Unlike other feature comparison methods, the RL aims to enhance the CD model's ability to detect building changes during the model fine-tuning stage. In Section III, we will provide comprehensive explanations of how our method is implemented.

III. Method

A. Framework Overview

Our proposed CD method is mainly based on a multitasking model. The model utilizes the semantic information from RS images effectively to detect change targets in paired images. The innovation is to combine the characteristics of image segmentation and CD and use the diffusion model to generate high-quality pseudo-change image pairs. These pseudo-change image pairs containing accurate change labels are the key to enhancing CD models.

We define the problem of transfer learning for CD as follows. Given a segmentation dataset of RS images $S^l = \{(x_i, y_i)\}_{i=1}^{M}$, where y_i is the semantic label of image x_i, and M denotes the total number of samples. Our goal is to enhance the CD performance by using the semantic information in RS images, we design the multitask network to perform both segmentation and CD. The model is trained in combination with an unlabeled CD dataset $D^{ul} = \{(I_1^n, I_2^n)\}_{n=1}^{N}$. I_1 and I_2 are RS images of different phases in the same area, and N is the number of unlabeled image pairs. The purpose of this approach is to train the multitask model using the labeled segmentation dataset S^l and the unlabeled change dataset D^{ul} and to improve the change prediction using semantic information.

The approach is mainly divided into three modules: GPM, which produces pseudo-change image pairs for the segmentation dataset of RS images based on DDPM; transfer learning model (TLM), which leverages semantic labels and pseudo-change labels simultaneously to improve the model's ability to detect change areas; RL is proposed to enhance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

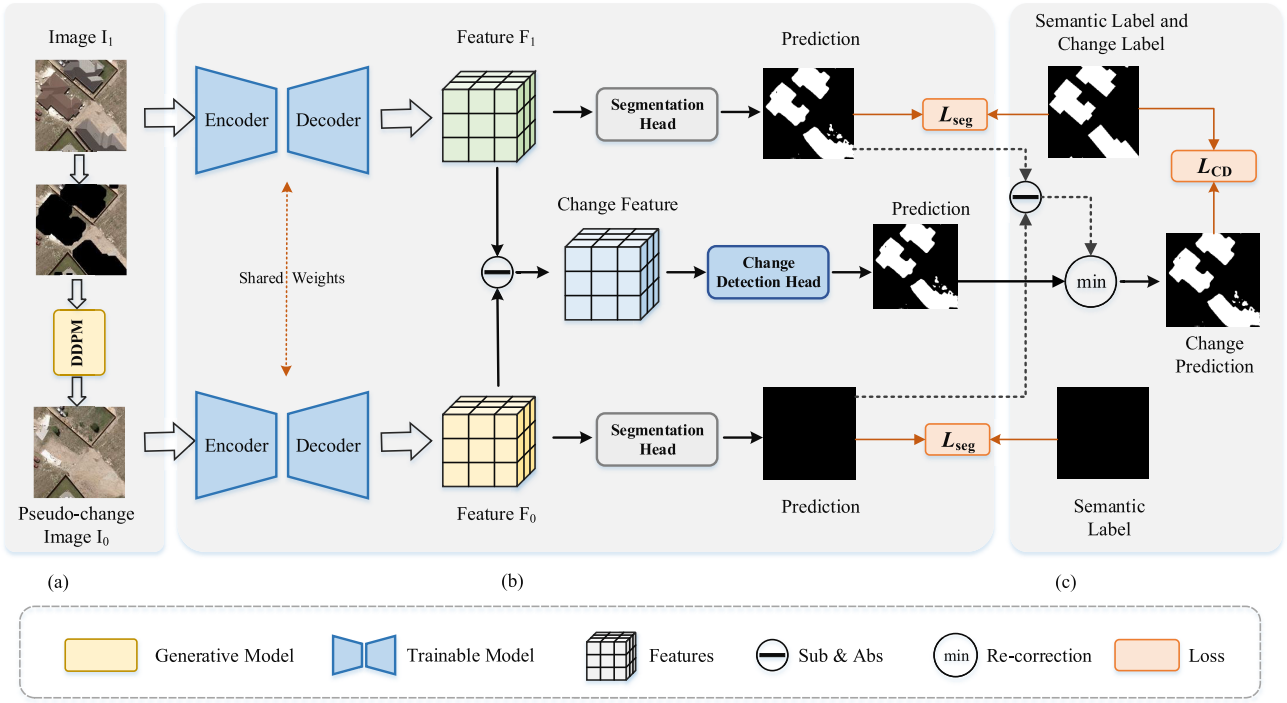


Fig. 2. Overview of transfer learning approach for CD. The original image I_1 is processed through DDPM to produce the pseudo-change image I_0, and then the segmentation results are obtained separately in the change prediction stage before the difference is computed. Finally, the pseudo-change labels and segmentation labels are utilized to guide the model's training. (a) GPM. (b) Change prediction. (c) Pseudo-change loss.

the discrimination ability of the model on the features of unchanged and changed regions.

B. Generation of Pseudo-Change Module

The main purpose of CD is to identify the changes of semantic objects between bi-temporal images. As shown in Fig. 2, in the proposed pseudo-change image generation method, we generate the change image by erasing the building objects in the original image under the guidance of semantic labels. Finally, pseudo-change image pairs generated by the DDPM model effectively improve the CD performance. Inspired by some recent image-denoising methods, we find that ddpm-based algorithms are usually able to generate more realistic images.

The process of generating pseudo-change image pairs is as follows: First, for the single-temporal RS image, we dilate the corresponding building semantic mask. Then, the building area was erased with the mask, the occluded image was repaired by the trained RS image diffusion generation model, and the simulated background scene was generated in the occluded area. Finally, by comparing this pair of images, it can be found that the original semantic labels have also become new pseudo-change labels for the training of CD models.

DDPM: Like other image generation methods, DDPM needs to learn the data distribution of RS images. To implement the pseudo-change image generation method, we initially train the diffusion model with the RS image dataset S^l. This ultimately produces RS image scenes that are more natural and realistic. During the training phase of the diffusion model, we initially apply random Gaussian noise $\mathcal{N}(0, I)$ at different t steps to the original RS image x_0 to obtain x_t. The model is then trained to generate the proper scene in the RS image. This process can be mathematically formulated as

$$q(x_t|x_0) = \mathcal{N}\left(x_t; \sqrt{\overline{\alpha}_t}x_0, (1-\overline{\alpha}_t)I\right) \tag{1}$$

where we calculate the noise variance as $\overline{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$. β_s is the variance for the Gaussian noise, and it is constantly changing with time t.

After training the DDPM, we obtain the diffusion model for generating random RS image scenes. To obtain image pairs of building changes, we use the building semantic label y of the image x as a mask m to erase buildings to create an image x' that does not contain buildings. However, directly using the semantic labels as masks leads to images that still have some building edges, which influence the generation of background regions. We implement image expansion on the original semantic labels m to obtain the erasure mask m'. The process of generating background regions using the diffusion model is defined as follows:

$$x' = \left(1 - m'\right) \odot x^{\text{back}} + m' \odot x^{pse}. \tag{2}$$

The background part of the image x^{back} comes from the area outside the mask m' in the original image. Then, in combination with the surrounding scene, x^{pse} is randomly generated by the diffusion model. An example of image generation is shown in Fig. 1.

Through the generation of pseudo-change images, we acquire RS images that comprise buildings I_1 in contrast to pseudo-change images that remove buildings I_0. To avoid the influence of buildings on the generation of backgrounds, we set a threshold γ for the selection of suitable images during the generation of pseudo-change images. The images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

utilized for pseudo-change image generation have a building area percentage below γ. In the multitask model training, we define the pseudo-change image pair (I_1, I_0) as the matrix X_{10}. Similarly, we use X_{12} to denote the matrix consisting of a pair of RS images without changing labels.

C. Transfer Learning-Based Model

After creating pseudo-change images, we can effectively transfer the semantic information in RS images to the change information between image pairs. To more effectively integrate semantic and change features in RS images, we develop a multitask transfer learning framework for detecting changes in RS data. As shown in Fig. 2, we obtain change image pairs generated by the GPM for model training. For the generated pseudo-change image pairs (I_1, I_0), their building labels y are also change labels. During the change prediction stage, the images I_1 and I_0 before and after the change are input into the feature extractor $F(\cdot)$ to obtain the corresponding features F_1 and F_0. The feature extraction models $F(\cdot)$ share parameters. To employ the semantic information of the images to guide the training, we use the same segmentation head $f(\cdot)$ to obtain the building object segmentation predictions for both images. The calculation of segmentation loss L_{seg} is defined as

$$L_{\text{seg}} = \mathcal{L}_{\text{seg}}(f(F_1), y) + \mathcal{L}_{seg}(f(F_0), y_0). \tag{3}$$

The loss $\mathcal{L}_{\text{seg}}$ is the sum of the binary cross-entropy loss and the dice coefficient loss. For the generated image I_0, we use the black label y_0 as the semantic label for training since the image does not contain any buildings.

Our change prediction approach differs from other methods that perform multiscale prediction at the feature extraction stage. We obtain change information separately at the feature and prediction levels. Finally, the model calculates absolute differences, which are then recorrected to produce the change prediction P_c

$$P_c = arg\min\{f(F_1) - f(F_0), f(F_1 - F_0)\}. \tag{4}$$

This change prediction at the end of the model has some advantages over other methods in the multitask framework. First, it combines semantic information and change information more effectively by comparing the differences in segmentation results and change prediction. This benefit is particularly useful for training multitask models. Second, this method also enables the feature extractor to be free from the interaction of multitasks, which makes it easier and more effective in training. The CD loss adopts the cross-entropy loss and the dice loss. The segmentation loss L_{seg} and change loss L_{CD} are combined, and the proposed multitask CD framework can be effectively trained to improve performance. The computation of loss on pseudo-change image pairs is defined as follows:

$$L(I_1, I_0) = L_{\text{seg}} + L_{CD}(P_c, y). \tag{5}$$

With the designed TLM, the proposed method allows the model to concentrate on the semantic features of the images and the change information between different images.

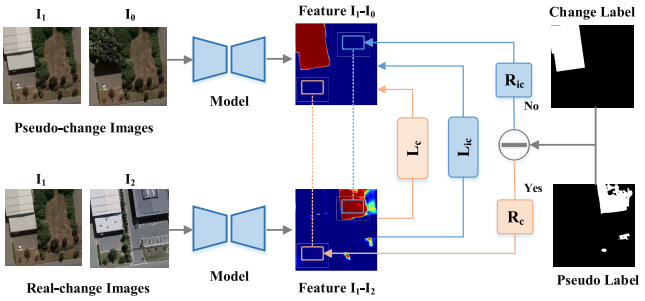


Fig. 3. Overview of the RL. The loss is calculated by comparing the change label with the pseudo-label to obtain the consistent mask R_c and the inconsistent mask R_{ic}, respectively.

D. Refined Loss

It should be noted that we have numerous pairs of real images without change labels. This is in addition to the generated pseudo-change images. To take full advantage of these unlabeled data, we initially generate pseudo-labels utilizing a pretrained model, similar to most semi-supervised techniques.

We propose the RL L_{ref} to improve the model's capacity for feature extraction in real building changes. Fig. 3 demonstrates that the pseudo-change image pair has a higher quality change label when compared to the two pairs of images. Furthermore, we employ the CycleGAN [18] as a style transfer model for both the remotely sensed segmentation dataset S^l and the unlabeled CD dataset D^{ul} to maintain consistent change data styles with the segmentation dataset, especially when there are significant style differences

$$X_{12} = F_{\text{trans}}\{X_{12} \to X_{10}\}. \tag{6}$$

The style transfer network efficiently transforms images from distinct datasets into a uniform scene style, consequently benefiting information transfer.

Notably, due to variances in label quality, the figure illustrates that the pseudo-change image pair better distinguishes the background in the edge region of the change object. The main purpose of the proposed RL is to utilize the pseudo-change prediction to guide the real change prediction for training. To achieve this, we first obtain consistent change mask $R_c = X(y_{10} = y_{12})$ and inconsistent change mask R_{ic} by comparing the change label with the pseudo-label. Then, we define a mask function $g(X, R)$ on image matrix X with mask region R, which outputs a masked image matrix of X based on mask R for further refined computation. Focusing on the region of change consistency, we apply the l_2 distance to put them closer

$$\begin{aligned} L_c &= \| \, g(P_c(X_{10}) - P_c(X_{12}), \; R_c) \, \|_2^2 \\ &= \frac{1}{M}\frac{1}{N}\sum_{i=1}^{M}\sum_{n=1}^{N}\left\| g\left(P_c\left(X_{10}^i\right) - P_c\left(X_{12}^n\right), \; R_c\right)\right\|_2^2. \end{aligned} \tag{7}$$

For regions with inconsistent changes, we predict the true change image pair opposite to the pseudo-change image pair. In loss L_{ic}, we instead compute the probability of invariance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF DIFFERENT RS DATASETS. NO CHANGE LABELS ARE USED FOR TRAINING ON ALL DATASETS, AND ONLY CHANGE LABELS ARE USED TO VERIFY MODEL PERFORMANCE DURING TESTING

Visual Problem	Datasets	Image Size	Train / Val / Test	Resolution	Semantic Label
Segmentation	WHU [4]	15354 × 32507	5948 / 743 / 743	0.2 m/pixel	✓
Change Detection	WHU-CD [4]	15354 × 32507	5948 / 743 / 743	0.2 m/pixel	✓
	LEVIR-CD [2]	1024 × 1024	7120 / 1024 / 2048	0.5 m/pixel	-
	GoogleGZ-CD [34]	1006 × 1168 - 4936 × 5224	2504 / 313 / 313	0.55 m/pixel	-

before pulling in their l_2 distance

$$L_{ic} = \| g(P_c(X_{10}) - |1 - P_c(X_{12})|, \ R_{ic}) \|_2^2 = \frac{1}{M}\frac{1}{N}\sum_{i=1}^{M}\sum_{n=1}^{N}\left\| g\left(P_c\left(X_{10}^i\right) - |1 - P_c\left(X_{12}^n\right)|, \ R_{ic}\right)\right\|_2^2. \tag{8}$$

Finally, the RL for change prediction is defined as

$$L_{\text{ref}} = \lambda L_c + L_{ic} \tag{9}$$

where λ is the weight of the change-consistent loss, which we set to 1.0 in our experiments.

Compared to other unsupervised CD and semi-supervised CD methods, the proposed transfer learning method utilizes the relation between semantic and change information of the RS image to solve different tasks and improve each other's performance. The training loss within the proposed transfer learning framework is defined as follows:

$$L_{\text{all}}(I_1, I_0, I_2) = \lambda_s L_{\text{seg}} + \lambda_{cd}\left(L_{CD} + L_{\text{CD-pse}}\right) + \lambda_{\text{ref}} L_{\text{ref}} \tag{10}$$

where L_{CD} and $L_{\text{CD-pse}}$ are also utilizing cross-entropy loss and dice loss, respectively. λ_s is the weight of segmentation loss and λ_{cd} is the weight of CD loss. Besides, λ_{ref} is the weight of RL. In our experiments, we usually set these weights to 1.0.

The proposed transfer learning approach involves pretraining the model with pseudo-change image pairs and then using the RL to improve the model. The model is finally used to effectively transfer the semantic information from the RS image dataset to the change prediction on extensive unlabeled CD data.

IV. EXPERIMENTS

A. Description of Datasets

To validate the effectiveness of the proposed transfer learning framework on RS images, we conducted extensive experiments on an RS image segmentation dataset and two RS image CD datasets. The WHU dataset [4] used in this work has both semantic and change labels, while the LEVIR-CD dataset [2] and the GoogleGZ-CD [34] only contain change labels. A comparison of the three datasets is shown in Table I.

The WHU dataset contains RS images of the same region in 2012 and 2016 with a size of 32 507 × 15 354 pixels. In addition, the resolution of the images is 0.2 m/pixel, and the semantic labels and change labels are buildings in the RS scene. Similar to other methods, we have cropped the original image pairs to obtain 7434 changed image pairs of size 256 × 256. During training, we split all images into 5948/743/743 image pairs for training/validation/testing. We trained our model using the respective semantic labels for each image pair as segmentation data. The proposed method is mainly concerned with verifying the performance improvement of semantic information for change prediction, so we only use the change labels in the validation and test phases.

For the WHU dataset, since it has both multitemporal change labels and single-temporal semantic labels, we mainly carried out the proposed pseudo-change generation method on this dataset and verified the effectiveness of the proposed method by real change labels. Among them, the segmentation label is mainly used to erase the building area and as the pseudo-change label after generating the pseudo-change image. The real change labels are used to validate the CD model.

The LEVIR-CD dataset consists of 637 high-resolution RS image pairs with a size of 1024 × 1024 pixels. The changes between different periods in the dataset mainly contain different types of buildings, such as houses, apartments, and garages. The spatial resolution of the images is 0.5 m/pixel, which is close to the WHU segmentation dataset we used. As a result, the dataset can serve as validation for the efficiency of the proposed transfer learning approach. In the experiment, we crop the original images to get images of size 256 × 256. Finally, 7120/1024/2048 pairs of images are used for model training, validation, and testing, respectively.

GoogleGZ-CD dataset includes 20 RS images of Guang Zhou at different time periods from 2006 to 2019. These images have great differences due to seasonal changes and water changes. The size of the images ranges from 1006 × 1168 to 4936 × 5224 with a spatial resolution of 0.55 m/pixel. To be suitable for the comparison experiments of different models, we crop the original images into 256 × 256 image blocks without overlapping. Finally, the data is divided into 2504/313/313 pairs of images are used for model training, validation, and testing, respectively.

B. Experimental Settings

The segmentation dataset is used to initially train the proposed network, and subsequently, the different temporal data are predicted separately to obtain the change prediction as the baseline. Meanwhile, we choose some recent CD methods as comparison algorithms: FC-EF [5], FC-Siam-con [5], FC-Siam-diff [5], DTCDSCN [40], and SEIFNet [32].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

To fairly compare the algorithm performance, we also use the pseudo-labels generated from the pretrained model of the RS segmentation dataset during training.

I3PE [44] is the latest method for single-temporal CD. In addition, we conduct a performance comparison between our proposed transfer learning method and the state-of-the-art transfer learning method, FFCTL [14]. This transfer method also uses segmentation datasets for training and then optimizes the CD model using pseudo-labels in multiple stages. The proposed network structure improves on FFCTL. Furthermore, we also compare the results of the generated pseudo-change images after training the CD model as a pretrained model (STCD†). We compare different methods on two RS images CD datasets, respectively.

All experiments are implemented in PyTorch with a single NVIDIA Quadro RTX 4080 GPU (16G memory). In our experiments, we use ResNet50 as the backbone of the feature extraction network. For multitemporal images, we employ encode and decode networks with shared parameters for feature extraction, respectively. The prediction is obtained after the segmentation head. We set the threshold γ to 0.5 when generating pseudo-change images based on the DDPM. To compare the effects of different thresholds, we also conduct experiments on the WHU-CD dataset. The consistent loss weight λ in the RL is set to 1.0. In all the comparison experiments, we train the model for 60 epochs, and the batch size is 4. We use the Adam optimizer with beta1 and beta2 set to 0.9 and 0.999, respectively. The basic learning rate for model training is set to $1e^{-3}$, and poly scheduling is employed to decay the learning rate. In the experiment of transferring the WHU dataset to the LEVIR-CD dataset, we use the CycleGAN network [18] to transfer image styles in the LEVIR-CD dataset.

C. Evaluation Criteria

As with other CD methods, we select five evaluation metrics in our experiments: intersection over union (IoU), $F1$ score, precision (Pre), recall (Rec), and overall accuracy (OA). IoU is commonly used to compare model performance in RS image segmentation and CD. In our experiments, we focus on the IoU results for change-building prediction. For both Pre and Rec metrics, we expect the model to produce high results. $F1$ score and OA offer a more comprehensive comparison of performance. It is worth noting that if the changed object represents a smaller section of the image, the OA may be lower due to the background's influence. These five evaluation metrics can be defined as

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}. \tag{11}$$

$$\mathrm{F1-score} = 2 \times \frac{\mathrm{Pre} \times \mathrm{Rec}}{\mathrm{Pre}+\mathrm{Rec}}. \tag{12}$$

$$\mathrm{Pre} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}. \tag{13}$$

$$\mathrm{Rec} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{14}$$

$$OA = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{TN}+\mathrm{FN}}. \tag{15}$$

TABLE II
RESULTS ON THE WHU-CD TEST SETS WITH DIFFERENT METHODS. THE RESULT WITH THE HIGHEST PRECISION IS MARKED IN RED, AND THE SECOND IS UNDERLINED

Method	IoU(%)	F1(%)	Pre(%)	Rec(%)	OA(%)
Baseline [45]	46.70	63.64	51.28	84.17	96.31
FC-EF [5]	57.64	73.13	78.47	68.46	97.85
FC-Siam-Diff [5]	56.97	72.58	75.97	69.48	97.96
FC-Siam-Conc [5]	60.03	75.03	83.81	67.91	98.24
DTCDSCN [40]	61.31	76.02	76.12	75.92	98.14
SEIFNet [32]	61.47	76.14	85.05	68.92	98.32
I3PE [44]	53.74	69.87	66.55	73.54	97.54
FFCTL [14]	60.37	75.29	79.14	71.71	98.17
STCD†	58.83	74.02	84.45	66.03	98.01
STCD	64.77	78.62	79.21	78.03	98.35

where true positive (TP) means that the model prediction at the pixel correctly identifies the area of change. False positive (FP) refers to the identification of irrelevant change as a change object by the model at the pixel. True negative (TN) indicates the correct prediction of the unchanged building area by the model. False negative (FN) refers to the misidentification of the change area as the unchanged class. To quantitatively compare various methods, we employed five metrics to evaluate them. The final results effectively compare the differences between the methods.

D. Quantitative Results

On the WHU dataset, as shown in Table II, the baseline results obtained by directly using segmentation to predict changes are relatively low for all metrics. Due to high segmentation prediction variance, a majority of the regions are identified as change targets, with a Rec of 84.17. The other CD models show significant performance improvement compared to the baseline since they use pseudo-change image pairs for pretraining. The latest CD model SEIFNet has also achieved good detection accuracy. Compared to FFCTL, which is also a transfer learning method, the proposed method can obtain the best change prediction. Compared with the other single-phase CD method I3PE, STCD also has significant performance improvement. We believe that its enhanced ability to convert semantic information into change information. Our approach attains an IoU increment of 4.4 percentage points, and it additionally achieves high prediction precision and recall rates.

Experiments on the LEVIR-CD dataset have demonstrated that the segmentation dataset WHU contains scenarios that are different from the CD dataset. Table III also shows that for different CD methods, the overall performance of the model drops significantly for all of them due to the lack of accurate change labels. The results demonstrate that FFCTL, a method that employs transfer learning, achieves remarkable performance improvement over other methods due to dataset variability. The baseline achieves the best OA results due to more background predictions, as the building changes in this RS scene are relatively sparse. However, summarizing the evaluation criteria, it can be seen that the transfer learning

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

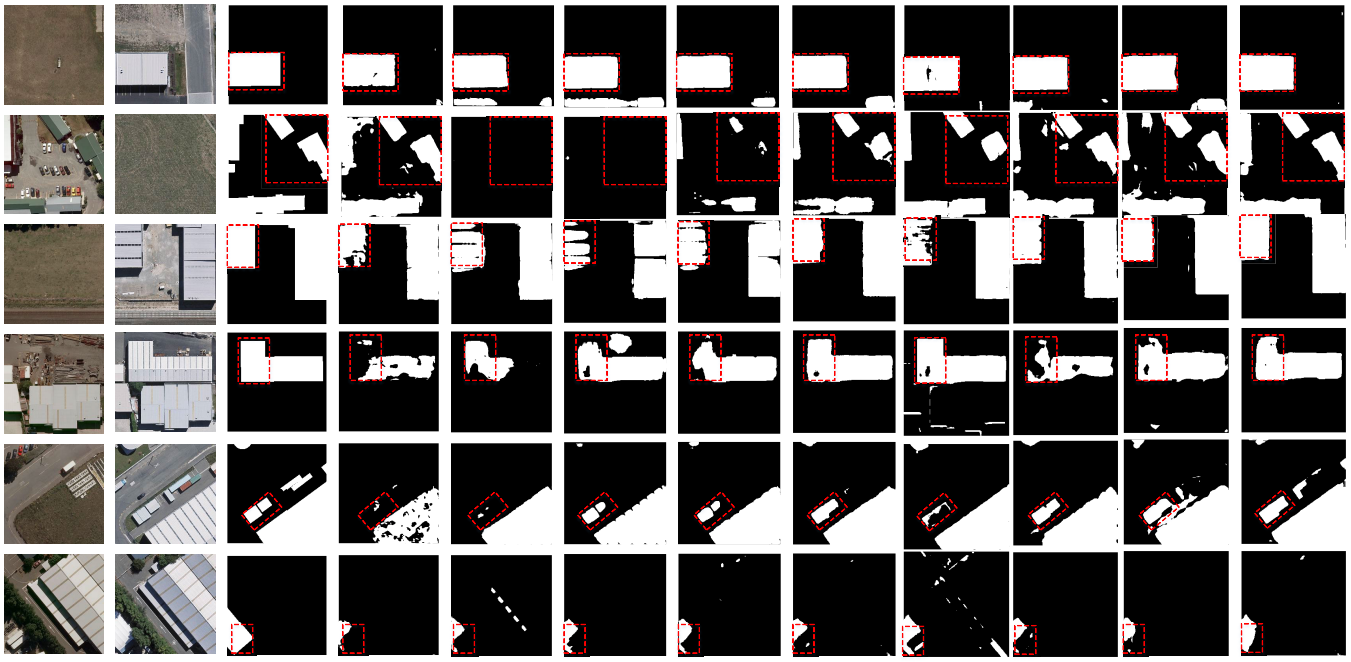


Fig. 4. Qualitative analysis results of CD on the WHU-CD test set. (a) Prechange images. (b) Postchange images. (c) Ground truth. (d) Baseline. (e) FC-EF. (f) FC-Siam-Diff. (g) FC-Siam-Conc. (h) DTCDSCN. (i) I3PE. (j) SEIFNet. (k) FFCTL. (l) STCD.

TABLE III

RESULTS ON THE LEVIR-CD TEST SETS WITH DIFFERENT METHODS. THE RESULT WITH THE HIGHEST PRECISION IS MARKED IN RED, AND THE SECOND IS UNDERLINED

Method	IoU(%)	F1(%)	Pre(%)	Rec(%)	OA(%)
Baseline [45]	9.06	16.61	34.27	10.96	95.32
FC-EF [5]	14.75	25.44	30.05	25.76	94.27
FC-Siam-Diff [5]	15.80	27.29	34.54	22.56	94.74
FC-Siam-Conc [5]	16.46	28.26	33.32	24.54	94.73
DTCDSCN [40]	16.61	28.49	29.66	27.40	94.23
SEIFNet [32]	18.51	31.24	31.01	31.54	94.28
I3PE [44]	10.77	19.45	28.44	14.78	90.12
FFCTL [14]	17.27	29.45	29.06	29.85	94.13
STCD†	16.32	28.05	27.70	28.41	93.88
STCD	21.57	35.49	36.73	34.33	94.76

TABLE IV

RESULTS ON THE GOOGLEGZ-CD TEST SETS WITH DIFFERENT METHODS. THE RESULT WITH THE HIGHEST PRECISION IS MARKED IN RED, AND THE SECOND IS UNDERLINED

Method	IoU(%)	F1(%)	Pre(%)	Rec(%)	OA(%)
Baseline [45]	19.60	32.71	75.8	20.9	92.6
FC-EF [5]	31.33	47.71	81.14	33.79	93.60
FC-Siam-Diff [5]	34.14	50.90	91.2	35.3	94.12
FC-Siam-Conc [5]	34.15	50.91	95.47	34.71	94.22
DTCDSCN [40]	32.19	48.70	87.04	33.81	93.85
SEIFNet [32]	34.56	51.37	74.96	39.07	93.61
I3PE [44]	31.77	48.22	87.83	33.23	93.84
FFCTL [14]	30.9	47.21	49.09	45.46	91.22
STCD†	27.90	43.62	57.54	35.28	92.25
STCD	37.23	54.26	65.06	41.34	93.02

method proposed in this article achieves the best change prediction performance. The differences between the different methods are also visible in the visualization of the prediction results of the model, with the proposed method being able to predict the area of change more accurately.

To verify the performance of the proposed method on more CD datasets, we supplement the comparative experiments on the GoogleGZ-CD dataset. The results on this dataset show that the latest model SEIFNet can achieve relatively good results. The results in Table IV show that the proposed STCD can effectively improve the accuracy of the CD model and has better results than other single-temporal label supervised training methods I3PE and FFCTL. Combining the experimental results of the three datasets, the proposed transfer learning method can use the pseudo-change images to enhance the recognition ability of the model for change information. The following sections will further analyze the experimental results by visualizing the results of some of the datasets.

E. Qualitative Analysis

Since the final goal of the proposed method is to accomplish the task of CD in RS images, we use the trained model for prediction on two test sets of RS image data, respectively. Fig. 4 presents the CD performance on the WHU dataset, which shows that the change prediction contains more irrelevant information when only the segmentation results are used. We have identified the most obvious areas of comparison with red boxes in the figure. It is also worth noting that due

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

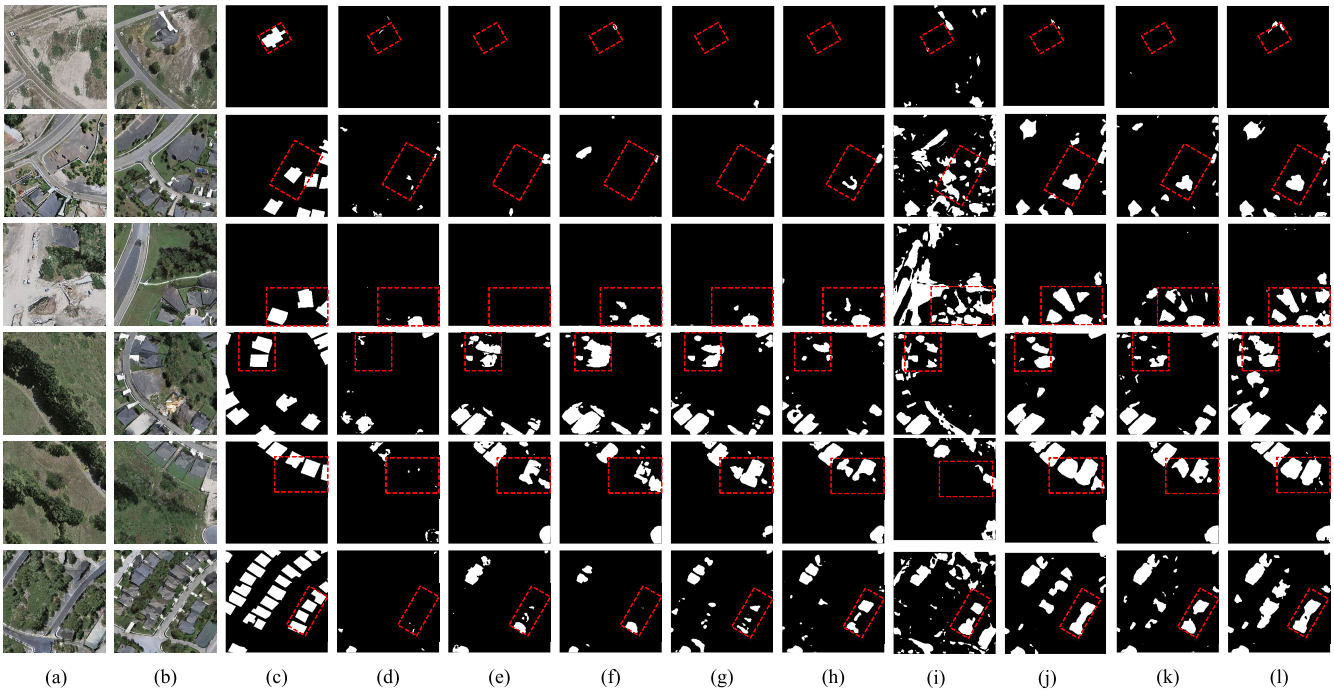


Fig. 5. Qualitative analysis results of CD on the LEVIR-CD test set. (a) Prechange images. (b) Postchange images. (c) Ground truth. (d) Baseline. (e) FC-EF. (f) FC-Siam-Diff. (g) FC-Siam-Conc. (h) DTCDSCN. (i) I3PE. (j) SEIFNet. (k) FFCTL. (l) STCD.

to the guided training of pseudo-change labels, the model's prediction of irrelevant changes also becomes more accurate, reducing false identifications. This also proves that the model can transfer semantic information into change information for better utilization.

When the segmentation dataset and the CD dataset have large differences, the prediction results of CD become rougher. Fig. 5 illustrates that for the LEVIR-CD dataset, most methods' predictions miss change regions when CD is performed across tasks and scenes. These results also show that the proposed method can better exploit the semantic information in images to improve CD performance when there is a large difference between the segmentation and CD data.

V. Discussion

A. Sensitivity of Parameter γ

In the proposed method, how to get more realistic and effective change image pairs is a key issue. In this approach, the threshold γ, which measures the percentage of building areas, is used to select some suitable images, and then the building areas in them are transformed into the background. To assess the impact of diverse threshold values (γ) on the transfer learning process, we analyzed the influence on CD performance using two CD datasets. Fig. 6 shows that a γ value of 0 does not result in the deletion of RS images to create pseudo-change image pairs, but relies exclusively on the segmentation model results to obtain change information, resulting in a decrease in model performance. As the threshold value γ increases incrementally, an increasing number of images are selected to create pseudo-change image pairs, leading to superior predictions for CD. Nevertheless, when γ is too large, some buildings are randomly generated in the

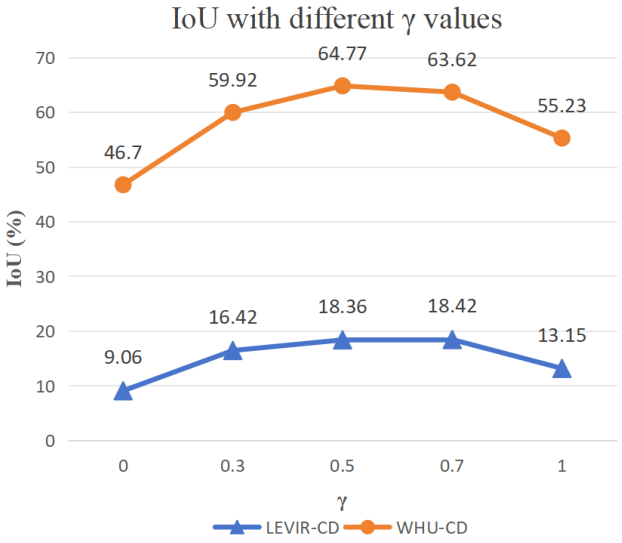
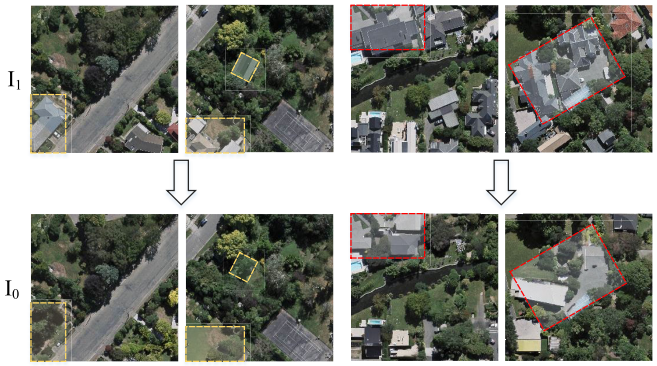


Fig. 6. Effect of threshold γ on the performance of CD models.

model-generated images due to the substantial percentage of building regions. Based on experimental analysis conducted on both datasets, we select 0.5 as the value of the threshold γ.

To better analyze the effect of threshold γ on pseudo-change image pairs, we select some generated pseudo-change image pairs for comparison. As shown in Fig. 7, when the building area is small in the image, the diffusion model can erase the buildings in it and generate more realistic background areas. However, when there are too many buildings in the image, it is easy to generate random other buildings to cover them because the generated region is affected by the surrounding scene. Our goal is to make change regions before and after erasure consistent with the original semantic labels, so this pseudo-change image makes the model predictions worse.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.



(a) High-quality Samples (b) Low-quality Samples

Fig. 7. Some samples of pseudo-change image generation: I_1 are original images and I_0 are the generated pseudo-change images. (a) Images of the building being successfully erased. (b) Generation of low-quality images.

In Fig. 7, when the threshold is large, the effect caused by these pseudo-change image pairs obtained from images containing more buildings causes the model performance to be worse. It can be demonstrated with these generated samples that we need to utilize the building percentage size to get more realistic and reliable pseudo-change images. In summary, when generating pseudo-change image pairs, it is important to select appropriate original images for a more effective improvement in CD performance.

In addition, since the proposed method and I3PE are similar in that they both use single-temporal semantic labels combined with images at $T1$ time to obtain pseudo-change images, we add some examples to show the differences between the two methods. As shown in Fig. 8, I3PE mainly divides the original $T1$ images into different small blocks and then randomly exchanges them to obtain pseudo-changed images. However, it can be seen from the example that the images after random exchange often have unreasonable splicing of background and foreground, which is also inconsistent with the change characteristics in natural scenes. In contrast, it can be seen from the samples in STCD that the main strategy of this method is to erase the original foreground buildings and then use the diffusion model to simulate the background to make the generated area naturally combine with the surrounding scene. The pseudo-change image obtained by this method is more natural, and the subsequent experimental results can also verify that the pseudo-change image pair can better strengthen the performance of the CD model.

B. Visual Analysis of RL

In the training phase of the model, we propose to utilize the RL to enable the prediction of unlabeled data to learn the prediction results of pseudo-change images. To better illustrate the RL's function across different image pairs, we visualize the prediction probability maps of some samples for comparison. As illustrated in Fig. 9, we generate the pseudo-change image pairs I_1 and I_0, whereas the unlabeled change image pairs in the CD dataset comprise I_1 and I_2. The probability distributions of their prediction demonstrate that the pseudo-change image pairs have semantic labels that can be used as change labels to guide model training, so their prediction probability maps have clearer predictions at the edges of the change objects. In contrast, because the unlabeled data is only trained with pseudo-labels to guide the training, more irrelevant regions in the prediction probability map are detected as change targets. Hence, the proposed RL in this article uses the blurred prediction of real change data to learn the accurate prediction features in the pseudo-change image pairs, and the TLM is finally improved in the feature recognition ability of change targets.

C. Ablation Experiments

In our transfer learning approach, we propose the generation algorithm of pseudo-change images, multitask model training, and model optimization with the RL. By combining these methods, the model ultimately achieves better CD performance. To analyze the impact of each module on the approach, we conduct ablation experiments using the WHU-CD dataset.

As shown in Table V, method (a) produces significantly lower accuracy than the other methods when the model is trained only with semantic labels. However, training the auxiliary multitask model with pseudo-change images obtained from the diffusion model leads to a significant improvement in the model's prediction accuracy for CD, with IoU increasing by 24 percentage points. In approach (c), the segmentation model is employed to acquire a pseudo-label optimization model for unlabeled change data. This not only enhances the model's performance but also provides some improvement. When we combine the pseudo-change images to train the multitask model and then use the model to acquire pseudo-labels to train the CD model, the change prediction IoU of the model has increased to 61.56.

Finally, during the training stage of the multitask model, we combine the characteristics of the unlabeled images and the pseudo-change images to further improve the performance of the model using the RL. The results of the ablation experiments prove that some important methods proposed in this article can effectively use semantic information to improve the performance of the CD model, and the combined method (e) can achieve the best CD accuracy.

To better compare the effect of RL with similar loss functions, we supplement the ablation experiments with a contrastive loss for CD. As a loss function in the semi-supervised CD method, contrastive loss can also strengthen the difference between image pairs. As can be seen from the table, the accuracy of the contrastive loss decreases slightly when the remaining methods are unchanged and only the RL is replaced. This result also proves the importance of RL in the proposed method, and it has better results than other loss functions.

D. Time Complexity Analysis

To evaluate the efficiency of the proposed TLM STCD, we evaluated its computational and space complexity with other recent CD models, including FC-EF, FC-Siam-Diff, FC-Siam-Conc, DTCDSCN, SEIFNet, and FFCTL in terms of floating-point operations per second (FLOPs) and the number of trainable parameters. As shown in Table VI, the early model is relatively simple, the amount of calculation and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

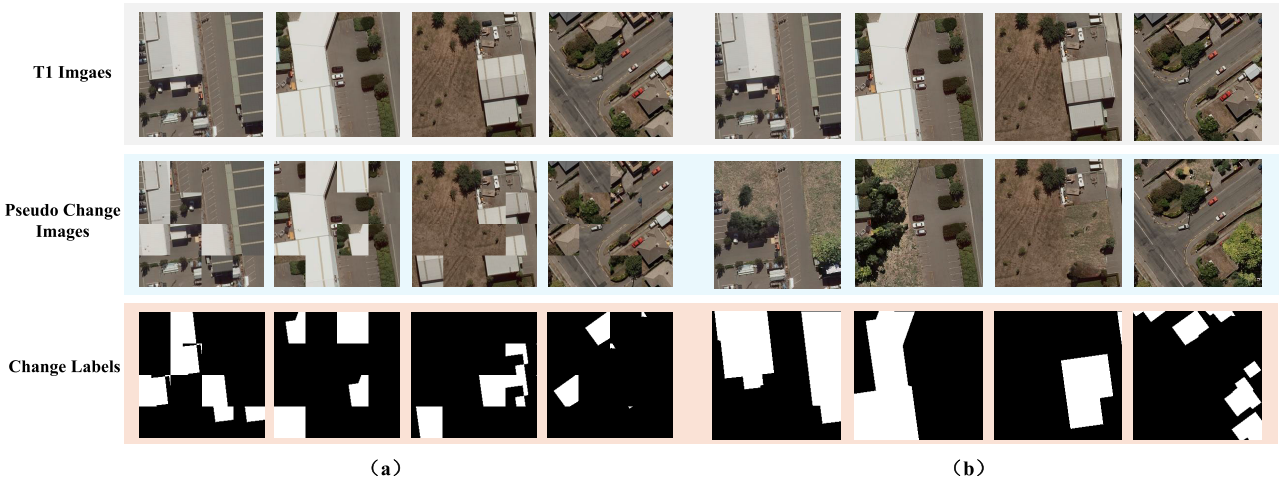


Fig. 8. Examples of the comparison between (a) I3PE and (b) STCD of pseudo-change image generation.

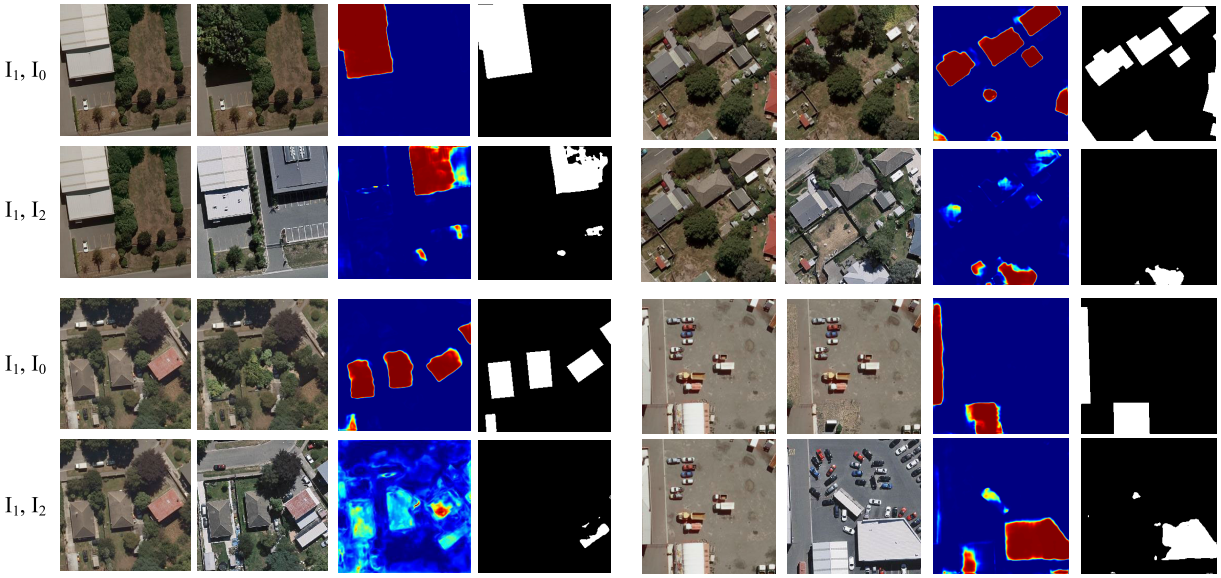


Fig. 9. Comparison of prediction probability maps for some pseudo-change image pairs and unlabeled image pairs.

TABLE V
EFFECTIVENESS OF DIFFERENT MODULES. THE RESULTS ARE COMPARED WITH THE IOU OF CHANGE AREAS

Methods	Pse-change	Pse-label	Contrastive Loss [46]	L_{ref}	IoU(%)
(a)					34.82
(b)	✓				58.35
(c)		✓			42.53
(d)	✓	✓			61.56
(e)	✓	✓	✓		63.21
(f)	✓	✓		✓	**64.77**

TABLE VI
COMPARISON OF COMPUTATION COMPLEXITY

Methods	Para (M)	GFLOPs (G)	IoU (%)
FC-EF [5]	1.35	3.57	57.64
FC-Siam-Diff [5]	1.35	4.72	56.97
FC-Siam-Conc [5]	1.55	5.32	60.03
DTCDSCN [40]	31.26	13.21	61.31
SEIFNet [32]	27.90	8.37	61.47
FFCTL [14]	32.52	26.72	60.37
STCD	32.52	21.42	**64.77**

parameters are low, but the accuracy of the CD model is relatively low. Compared with the models of other transfer learning methods, the proposed method has no increase in the number of parameters, but the performance of the model is better while the calculation is reduced. This indicates that the proposed pseudo-change label has a more effective role, while the refining loss not only does not increase the computational complexity but also further improves the detection accuracy of the model for the changed region.

E. Limitations and Future Work

In general, the proposed method in this article can more effectively utilize the semantic information in RS images to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

improve the CD prediction accuracy. However, the method also has some limitations due to its complexity. In the pseudo-label generation method, both the semantic target and the background change in different RS scenes. Therefore, during the DDPM training, the model initially familiarizes itself with the generation distribution of RS images based on various scenes. This necessitates retraining of the diffusion model when the semantic segmentation dataset of RS images is changed. Besides, the generation of images using DDPM is relatively time-consuming.

In future studies, we aim to explore the direct training of generative models for controllable classes using semantic labels on different RS datasets. This allows for the transfer of semantic information from images to change information via image generation methods, resulting in an improved effectiveness of multitask model training.

VI. Conclusion

The CD labels of RS images are usually more difficult to obtain than the semantic labels, so how to transfer the semantic information into change information deserves to be explored. In this article, we propose a transfer learning method based on pseudo-change image generation, which utilizes a multitask network to simultaneously achieve the tasks of semantic segmentation and CD in RS images. In addition, the model combines pseudo-change image pairs and unlabeled image pairs in the model training process, and the model uses the RL to make the change features have better discriminative ability.

To validate the effectiveness of the proposed method, we conduct a series of experiments. The experimental results on the WHU and LEVIR-CD datasets demonstrate that the proposed method outperforms the latest transfer learning methods. Furthermore, the results of generation samples of pseudo-change images and ablation experiments verified the effect of each module in the method, respectively. The proposed transfer learning approach leverages the semantic information of RS images and offers new solution ideas for the practical application of CD.

Acknowledgment

The authors would like to thank the editor and anonymous reviewers for their valuable comments and suggestions, which are very helpful in improving this article.

References

[1] Y. Xu, L. Wu, Z. Xie, and Z. Chen, "Building extraction in very high resolution remote sensing imagery using deep learning and guided filters," *Remote Sens.*, vol. 10, no. 1, p. 144, Jan. 2018.

[2] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[3] J. Z. Xu, W. Lu, Z. Li, P. Khaitan, and V. Zaytseva, "Building damage detection in satellite imagery using convolutional neural networks," 2019, *arXiv:1910.06444*.

[4] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.

[5] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[6] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[7] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2006.

[8] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, 1998.

[9] W. Gedara Chaminda Bandara and V. M. Patel, "Deep metric learning for unsupervised remote sensing change detection," 2023, *arXiv:2303.09536*.

[10] S. W. Zamir et al., "ISAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.

[11] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Can semantic labeling methods generalize to any city? The inria aerial image labeling benchmark," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 3226–3229.

[12] F. Rottensteiner et al., "The ISPRS benchmark on urban object classification and 3D building reconstruction," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.; I-3*, vol. 1, no. 1, pp. 293–298, 2012.

[13] H. Zunair, Y. Gobeil, S. Mercier, and A. B. Hamza, "STAR: Noisy semi-supervised transfer learning for visual classification," in *Proc. 4th Int. Workshop Multimedia Content Anal. Sports*, Oct. 2021, pp. 25–33.

[14] Y. Cao and X. Huang, "A full-level fused cross-task transfer learning method for building change detection using noise-robust pretrained networks on crowdsourced labels," *Remote Sens. Environ.*, vol. 284, Jan. 2023, Art. no. 113371.

[15] A. Q. Nichol and P. Dhariwal, "Improved denoising diffusion probabilistic models," in *Proc. 38th Int. Conf. Mach. Learn.*, vol. 139, Jul. 2021, pp. 8162–8171.

[16] A. Lugmayr, M. Danelljan, A. Romero, F. Yu, R. Timofte, and L. van Gool, "RePaint: Inpainting using denoising diffusion probabilistic models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11461–11471.

[17] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. NIPS*, vol. 34, 2021, pp. 8780–8794.

[18] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2223–2232.

[19] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, no. 1, pp. 1–14, Apr. 2021.

[20] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[21] Y. Zhang, H. Zhang, N. M. Nasrabadi, and T. S. Huang, "Multi-metric learning for multi-sensor fusion based classification," *Inf. Fusion*, vol. 14, no. 4, pp. 431–440, Oct. 2013.

[22] Q. Yan et al., "Uncertainty estimation in HDR imaging with Bayesian neural networks," *Pattern Recognit.*, vol. 156, Dec. 2024, Art. no. 110802.

[23] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–14.

[24] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[25] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, no. 2, pp. 1–5, Apr. 2021, doi: 10.1109/LGRS.2021.3056416.

[26] H. Zhang, G. Ma, Y. Zhang, B. Wang, H. Li, and L. Fan, "MCHA-Net: A multi-end composite higher-order attention network guided with hierarchical supervised signal for high-resolution remote sensing image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 202, pp. 40–68, Aug. 2023.

[27] R. Zhang, H. Zhang, X. Ning, X. Huang, J. Wang, and W. Cui, "Global-aware Siamese network for change detection on remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 199, pp. 61–72, May 2023.

[28] S. Saha, L. Mou, X. X. Zhu, F. Bovolo, and L. Bruzzone, "Semisupervised change detection using graph convolutional network," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 4, pp. 607–611, Apr. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.

[29] X. Zheng, P. Li, X. Hu, and K. Yu, "Semi-supervised classification on data streams with recurring concept drift and concept evolution," *Knowl.-Based Syst.*, vol. 215, Mar. 2021, Art. no. 106749.
[30] W. Zhang, X. Lu, and X. Li, "A coarse-to-fine semi-supervised change detection for multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3587–3599, Jun. 2018.
[31] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.
[32] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414, doi: 10.1109/TGRS.2024.3360516.
[33] I. Goodfellow et al., "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 27, 2014, pp. 1–15.
[34] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.
[35] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, vol. 33, 2020, pp. 6840–6851.
[36] Q. Yan et al., "Toward high-quality HDR deghosting with conditional diffusion models," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 4011–4026, May 2024.
[37] L. Lu, J. Li, J. Cao, L. Niu, and L. Zhang, "Painterly image harmonization using diffusion model," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 233–241.
[38] Y. Han, J. He, M. Gupta, E. J. Delp, and F. Zhu, "Diffusion model with clustering-based conditioning for food image generation," in *Proc. 8th Int. Workshop Multimedia Assist. Dietary Manage.*, vol. 32, Oct. 2023, pp. 61–69.
[39] H. Noh, J. Ju, M. Seo, J. Park, and D.-G. Choi, "Unsupervised change detection based on image reconstruction loss," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 1351–1360.
[40] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.
[41] J. Liu et al., "Convolutional neural network-based transfer learning for optical aerial images change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 1, pp. 127–131, Jan. 2020.
[42] T. Liu et al., "Building change detection for VHR remote sensing images via Local–Global pyramid network and cross-task transfer learning strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4704817, doi: 10.1109/TGRS.2021.3130940.
[43] S. Saha, Y. T. Solano-Correa, F. Bovolo, and L. Bruzzone, "Unsupervised deep transfer learning-based change detection for HR multispectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 856–860, May 2021.
[44] H. Chen, J. Song, C. Wu, B. Du, and N. Yokoya, "Exchange means change: An unsupervised single-temporal change detection framework based on intra- and inter-image patch exchange," *ISPRS J. Photogramm. Remote Sens.*, vol. 206, pp. 87–105, Dec. 2023.
[45] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[46] J.-X. Wang, T. Li, S.-B. Chen, J. Tang, B. Luo, and R. C. Wilson, "Reliable contrastive learning for semi-supervised change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4416413.



Jia-Xin Wang received the B.S. degree from Hefei University, Hefei, China, in 2016, and the Ph.D. degree in computer science from Anhui University of China, Hefei, in 2021.

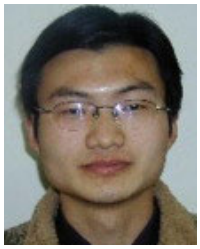
From 2022 to 2024, he was a Post-Doctoral Researcher with Anhui University. He is currently an Assistant Professor with Anhui University of Science and Technology, Hefei. His research interests include computer vision, deep learning, and remote sensing.



Teng Li received B.S. degree from the University of Science and Technology of China (USTC), Hefei, China, in 2001, the M.S. degree from the Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing, China, in 2004, and the Ph.D. degree from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 2010.

He is currently a Tenure-Track Professor with Anhui University, Hefei.

Dr. Li received the IEEE T-CSVT Best Paper award in 2014 and the ICIMCS Best Paper Award in 2009.



Si-Bao Chen (Member, IEEE) received the B.S. and M.S. degrees in probability and statistics and the Ph.D. degree in computer science from Anhui University of China, Hefei, China, in 2000, 2003, and 2006, respectively.

From 2006 to 2008, he was a Post-Doctoral Researcher with the Department of Electronic Engineering and Information Science, University of Science and Technology of China, Hefei. Since 2008, he has been a Teacher with Anhui University. From 2014 to 2015, he was a Visiting Scholar with the University of Texas at Arlington, Arlington, TX, USA. His research interests include image processing, pattern recognition, machine learning, and computer vision.



Cheng-Jie Gu received the Ph.D. degree from Nanjing University of Posts and Telecommunications, Nanjing, China, in 2012.

From 2012 to 2017, he was an Innovation Team Leader with the 38th Research Institute of CETC and conducted research and development in the communication and networking sector. He has completed Post-Doctoral Research with the USTC. He is a High-Level Innovation Leader in Anhui province and a Cybersecurity Expert in Zhejiang province, China. Currently, he is a Dean of the School of Public Security and Emergency, Anhui University of Science and Technology, Hefei, China. His research interest includes information network and trusted network architecture.



Zhi-Hui You received the B.S. degree in computer science from Jiangxi Agricultural University, Nanchang, China, in 2020. He is currently pursuing the Ph.D. degree in computer science with Anhui University of China, Hefei, China.

His research interests include computer vision, pattern recognition, semantic segmentation, and change detection in remote sensing.



Bin Luo (Senior Member, IEEE) received the B.Eng. degree in electronics and the M.Eng. degree in computer science from Anhui University, Hefei, China, in 1984 and 1991, respectively, and the Ph.D. degree in computer science from the University of York, York, U.K., in 2002.

From 2000 to 2004, he was a Research Associate with the University of York. He is currently a Professor with Anhui University. His research interests include graph spectral analysis, large image database retrieval, image and graph matching, statistical pattern recognition, digital watermarking, and information security.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:14:23 UTC from IEEE Xplore. Restrictions apply.